In [50]:
```
# 1. Read a file containing student names(one per line) and store them in a list.

with open("students.txt", "r") as file:
    names = file.read().splitlines()

print("Student names: ", names)
```

```
Student names:  ['Tanvi Jagdish Khedekar', 'Gauri Jagdish Khedekar', 'Rani', 'Vaishn
avi Chandan', 'Saurabh Mangrulkar', 'Gauri Jagdish Khedekar', 'Raj', 'Bhumicka Desa
i', 'Om Deo', 'Manu', 'Anu']
```

In [51]:
```
# 2. Print names longer that 5 characters.

with open("students.txt", "r") as file:
    names = file.read().splitlines()

print("Student name longer than 5 characters: ")

for name in names:
    if len(name) > 5:
        print(name)
```

```
Student name longer than 5 characters:
Tanvi Jagdish Khedekar
Gauri Jagdish Khedekar
Vaishnavi Chandan
Saurabh Mangrulkar
Gauri Jagdish Khedekar
Bhumicka Desai
Om Deo
```

In [52]:
```
# 3. Count how many names start with a vowel.

with open("students.txt", "r") as file:
    names = file.read().splitlines()

count = 0
for name in names:
    if name[0].lower() in "aeiouAEIOU":
        count += 1

print("Number of names starting with a vowel: ", count)
```

```
Number of names starting with a vowel:  2
```

In [53]:
```
# 4. Save the filtered names(longer than 5 characters) into a new file.

with open("students.txt", "r") as file:
    names = file.read().splitlines()

filtered_names = []
for name in names:
    if len(name) > 5:
        filtered_names.append(name)
```

```
with open("filtered_names.txt", "w") as file:
    for name in filtered_names:
        file.write(name + "\n")

print("Filtered names saved successfully!")
print(filtered_names)
```

```
Filtered names saved successfully!
['Tanvi Jagdish Khedekar', 'Gauri Jagdish Khedekar', 'Vaishnavi Chandan', 'Saurabh M
angrulkar', 'Gauri Jagdish Khedekar', 'Bhumicka Desai', 'Om Deo']
```

In [54]:
```
# 5. Sort the list of names alphabetically and print them in ascending order.

with open("students.txt", "r") as file:
    names = file.read().splitlines()

names.sort()
print("Names in ascending order: ")

for name in names:
    print(name)
```

```
Names in ascending order:
Anu
Bhumicka Desai
Gauri Jagdish Khedekar
Gauri Jagdish Khedekar
Manu
Om Deo
Raj
Rani
Saurabh Mangrulkar
Tanvi Jagdish Khedekar
Vaishnavi Chandan
```

In [55]:
```
# 6. Print names that contain the letter "a".

with open("students.txt", "r") as file:
    names = file.read().splitlines()

print("Names that contain the letter 'a': ")
for name in names:
    if "a" in name.lower():
        print(name)
```

```
Names that contain the letter 'a':
Tanvi Jagdish Khedekar
Gauri Jagdish Khedekar
Rani
Vaishnavi Chandan
Saurabh Mangrulkar
Gauri Jagdish Khedekar
Raj
Bhumicka Desai
Manu
Anu
```

In [39]:
```python
# 7. Count word frequency in names (handle duplicates) and store results in a dicti

with open("students.txt", "r") as file:
    names = file.read().splitlines()

word_frequency = {}

for name in names:
    if name in word_frequency:
        word_frequency[name] += 1
    else:
        word_frequency[name] = 1

print("Name frequency: ")
print(word_frequency)
```

```
Name frequency:
{'Tanvi Jagdish Khedekar ': 1, 'Gauri Jagdish Khedekar': 2, 'Rani': 1, 'Vaishnavi Ch
andan': 1, 'Saurabh Mangrulkar': 1, 'Raj': 1, 'Bhumicka Desai': 1, 'Om Deo': 1, 'Man
u': 1, 'Anu': 1}
```

In [57]:
```python
# 8. String analysis - for each name, print: length of name, first and last charact

with open("students.txt", "r") as file:
    names = file.read().splitlines()

for name in names:
    print("Name: ", name)
    print("Length of the name: ", len(name))
    print("First character: ", name[0])
    print("Last character: ", name[-1])
    if name == name[::-1]:
        print("This is a Palindrome.")
    else:
        print("This is not a Palindrome.")
    print("\n")
```

```
Name:  Tanvi Jagdish Khedekar
Length of the name:  22
First character:  T
Last character:  r
This is not a Palindrome.


Name:  Gauri Jagdish Khedekar
Length of the name:  22
First character:  G
Last character:  r
This is not a Palindrome.


Name:  Rani
Length of the name:  4
First character:  R
Last character:  i
This is not a Palindrome.


Name:  Vaishnavi Chandan
Length of the name:  17
First character:  V
Last character:  n
This is not a Palindrome.


Name:  Saurabh Mangrulkar
Length of the name:  18
First character:  S
Last character:  r
This is not a Palindrome.


Name:  Gauri Jagdish Khedekar
Length of the name:  22
First character:  G
Last character:  r
This is not a Palindrome.


Name:  Raj
Length of the name:  3
First character:  R
Last character:  j
This is not a Palindrome.


Name:  Bhumicka Desai
Length of the name:  14
First character:  B
Last character:  i
This is not a Palindrome.
```

```
Name:  Om Deo
Length of the name:  6
First character:  O
Last character:  o
This is not a Palindrome.


Name:  Manu
Length of the name:  4
First character:  M
Last character:  u
This is not a Palindrome.


Name:  Anu
Length of the name:  3
First character:  A
Last character:  u
This is not a Palindrome.
```

In [70]:
```python
# 9 . File + List + String Combined Challenge:
# Read names from file -> list.
# Remove duplicates.
# Sort by length (shortest to longest).
# Write results into processed_names.txt

with open("students.txt", "r") as file:
    names = file.read().splitlines()

processed_name = list(set(names))  # lists don't store duplicate values
processed_name.sort(key = len)   # using key=length to sort the list

with open("processed_names.txt", "w") as file:
    for name in processed_name:
        file.write(name + "\n")   # writing into new file
print("Duplicates removed successfully!\n")
print("Names sorted by length:")
print(processed_name)
print("\nResults saved in 'processed_names.txt'")
```

```
Duplicates removed successfully!

Names sorted by length:
['Raj', 'Anu', 'Manu', 'Rani', 'Om Deo', 'Bhumicka Desai', 'Vaishnavi Chandan', 'Sau
rabh Mangrulkar', 'Tanvi Jagdish Khedekar', 'Gauri Jagdish Khedekar']

Results saved in 'processed_names.txt'
```

In [85]:
```python
# 10. Read names from file.
# Group them by starting letter.
# Save each group into separate files.

with open("students.txt", "r") as file:
    names = file.read().splitlines()
```

```
    for _ in "abcdefghijklmnopqrstuvwxyz":
        for name in names:
            letter = name[0].lower()
            with open(letter + "_names.txt", "w") as file:
                file.write(name)
print("All files created successfully!")
```

```
All files created successfully!
```

In [ ]: